

1. Define Data Mining and list any two basic data mining tasks. Explain the KDD process and the role of data preprocessing.

Ans) • Data Mining:

The process of discovering useful patterns and knowledge from large datasets.

• Two Tasks: classification, clustering.

• KDD steps: Data cleaning → Data Integration → Data selection → Data Transformation → Data Mining → Pattern Evaluation → knowledge presentation.

• Role of preprocessing: Improves data quality by removing errors, missing values, and inconsistencies, leading to accurate mining results.

2. What is data cleaning? Mention two data quality issues.

Explaining methods to handle missing values and noisy data.

Ans)

• Data cleaning: process of removing errors and inconsistencies from data.

• Two issues: Missing values, Noisy Data.

• Handling Missing values: Replace with mean/median or delete records.

• Handling Noisy Data: Use binning or smoothing.

• Example: Missing age → replace with average age

3) Define data integration and list two challenges. Explain handling redundancy and inconsistency.

Ans. Data Integration: combining data from multiple sources into one dataset.

- challenges: Redundancy, Data Inconsistency.

- Redundancy: Remove duplicate records.

- Inconsistency: standardize formats and resolve conflicts.

4) What is covariance analysis? list the two covariance formulas.

Ans) Covariance Analysis: Measures how two variables change together.

- Formulas:

a. population covariance: $\text{cov}(x, y) = \sum [(x_i - \bar{x})(y_i - \bar{y})] / N$

b. Sample covariance: $\text{cov}(x, y) = \sum [(x_i - \bar{x})(y_i - \bar{y})] / (n - 1)$

5) Stocks A and B values: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Will they rise or fall together?

Ans) The values of both stocks generally increase together.

- covariance is positive

- Therefore, stock A and stock B rise together, not independent.

6) Explain the architecture of a typical data mining system.

Ans) Architecture:

Database



Database/Data Warehouse



Data cleaning & Integration



Data Mining Engine



Pattern Evaluation



Knowledge User Interface

7) Find covariance.

Given:

• $x = \{4, 5, 6, 7, 9\}$

• $y = \{8, 2, 7, 5, 6\}$

Answer:

• $\text{Mean}(x) = 6.2$

• $\text{Mean}(y) = 5.8$

• $\text{Covariance} = -0.48$

• Negative covariance indicates an inverse relationship.

8) Describe major steps in data preprocessing and suggest a preprocessing pipeline.

Ans) Steps:

1. Data cleaning.

2. Data Integration

3. Data Transformation

4. Data Reduction

Pipeline: Missing values $\rightarrow$ Remove outliers $\rightarrow$ standardize formats $\rightarrow$ Normalize data.

This sequence improves data quality before mining.

9) Perform equal-frequency (equal-depth) binning for the given age data.

Ans Total values = 27

Divide into 3 equal-frequency bins (9 values each):

• Bin1: 13, 15, 16, 16, 19, 20, 20, 21, 22

• Bin2: 22, 25, 25, 25, 25, 30, 33, 35

• Bin3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Equal-frequency binning places an equal number of values in each bin.

10) Explain data discretization. compare equal-width and equal-frequency binning

Ans • Data Discretization: converts continuous data into intervals.

• Importance: Reduces noises and simplifies analysis.

Equal-width

Equal-frequency.

Equal Interval size

Equal number of values

Simple to implement

Handles skewed data better

Ex:- 0-10 to 10-20

Each bin has same number of records